



Ansys Fluent Simulation Report

Analyst	Thinkpad
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System Information

Application	Fluent
Settings	3d, pressure-based, SST k-omega
Version	24.1.0-10184
Source Revision	5b3f9fb3c8
Build Time	Nov 22 2023 10:25:54 EST
CPU	Intel(R) Core(TM) i7-7500U
OS	Windows

Geometry and Mesh

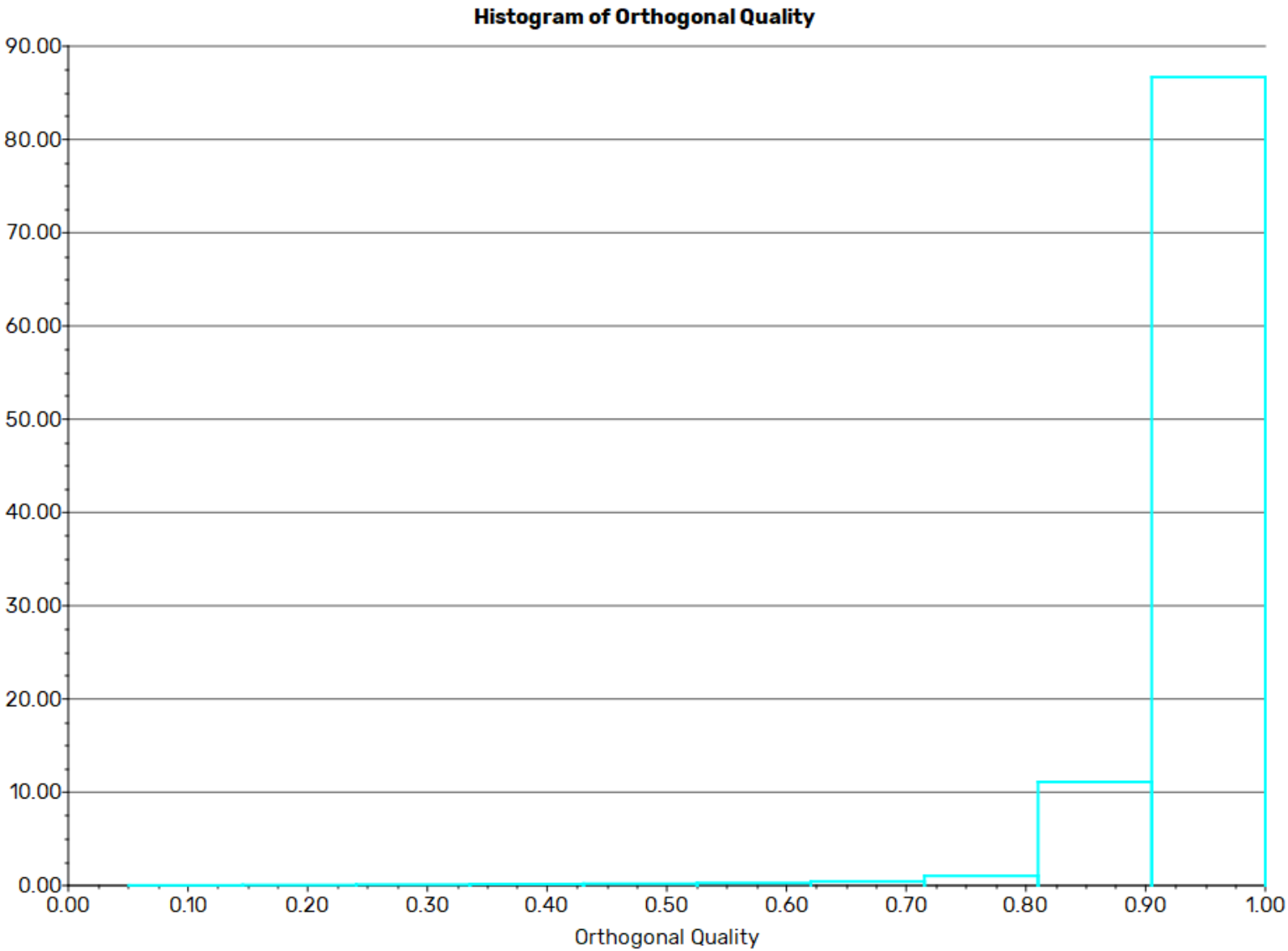
Mesh Size

Cells	Faces	Nodes
1291222	5568328	3118737

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fluid	Mixed Cell	0.050778031	264.59344

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	SST k-omega turbulence model
Heat Transfer	Enabled

Material Properties

— Fluid	
— air	
Density	ideal gas
Cp (Specific Heat)	1006.43 J/(kg K)
Thermal Conductivity	0.0242 W/(m K)
Viscosity	sutherland
Molecular Weight	28.966 kg/kmol
— Solid	
— aluminum	
Density	2719 kg/m^3
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)

Cell Zone Conditions

— Fluid	
— fluid	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no
3D Fan Zone?	no

Boundary Conditions

— Inlet	
— pressure-farfield	
Gauge Pressure [Pa]	0

Mach Number	0.7
Temperature [K]	300
Coordinate System	Cartesian (X, Y, Z)
Component of Flow Direction (x,y,z)	(1, 0, 0)
Turbulent Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
— Symmetry	
fluid-symmetry	symmetry
— Wall	
— wing	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	rough bc standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1
— fluid:1	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	rough bc standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1
— wing-edge	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip

Wall Surface Roughness	rough bc standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1

Reference Values

Area	1 m^2
Density	1.176655 kg/m^3
Enthalpy	31377.64 J/kg
Length	1 m
Pressure	0 Pa
Temperature	299.9999 K
Velocity	242.9639 m/s
Viscosity	1.846179e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fluid

Solver Settings

— Equations	
Flow	True
Turbulence	True
Energy	True
— Numerics	
Absolute Velocity Formulation	True
— Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1
Turbulent Kinetic Energy	0.75
Specific Dissipation Rate	0.75
Turbulent Viscosity	1
Energy	0.75
Explicit Momentum	0.5
Explicit Pressure	0.5
— Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
— Discretization Scheme	
Pressure	Second Order
Density	Second Order Upwind
Momentum	Second Order Upwind
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind

Energy	Second Order Upwind
— Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Spec. Dissipation Rate [s^-1]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	4
Case Read	14.534 seconds
Iteration	2117.8 seconds
AMG	1460.49 seconds
Virtual Current Memory	4.71905 GB
Virtual Peak Memory	5.73323 GB
Memory Per M Cell	3.4121

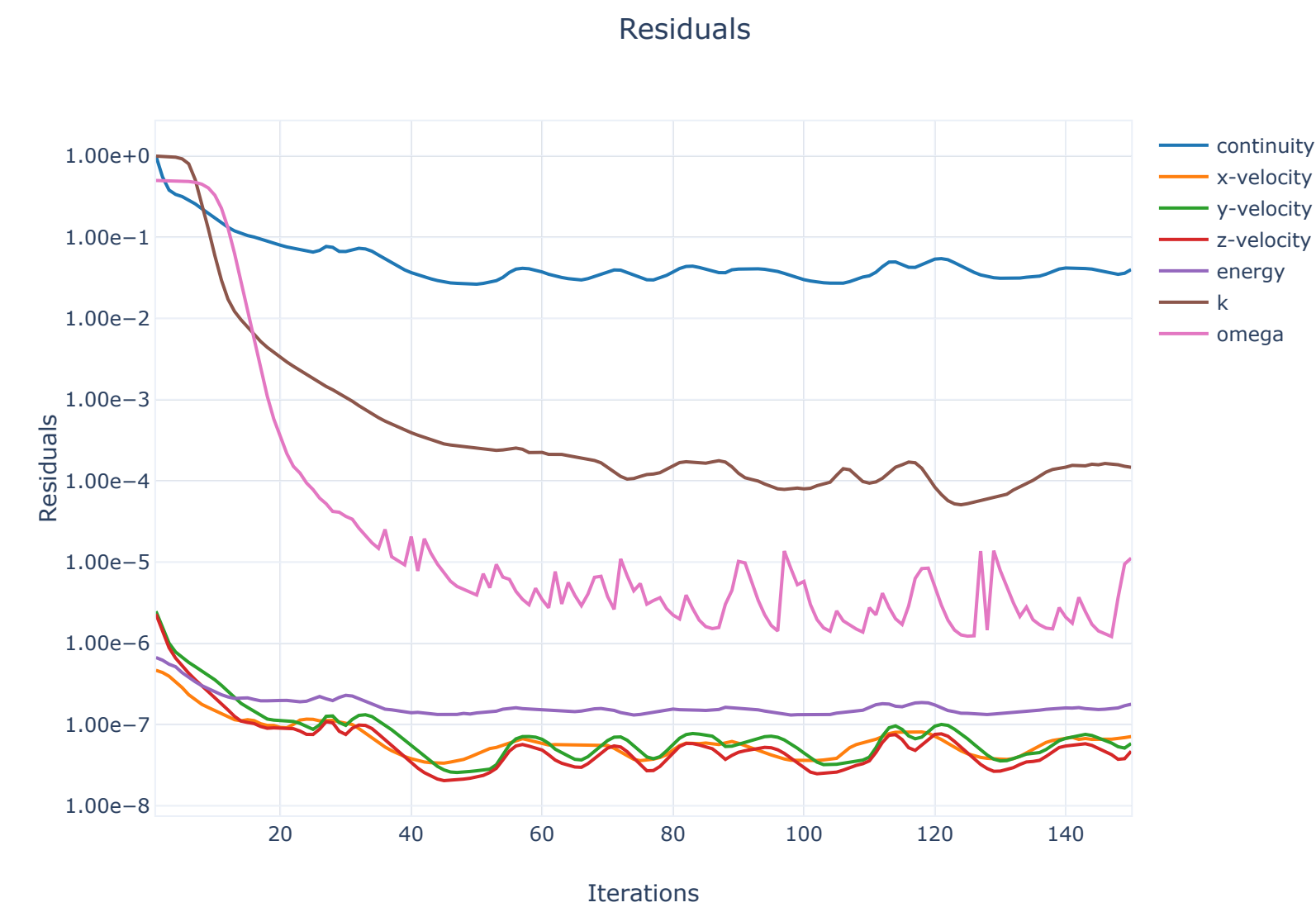
Solution Status

Iterations: 150

	Value	Absolute Criteria	Convergence Status
continuity	0.03995666	0.001	Not Converged
x-velocity	7.124125e-08	0.001	Converged
y-velocity	5.849519e-08	0.001	Converged
z-velocity	4.701925e-08	0.001	Converged
energy	1.782164e-07	1e-06	Converged
k	0.0001470588	0.001	Converged
omega	1.121795e-05	0.001	Converged

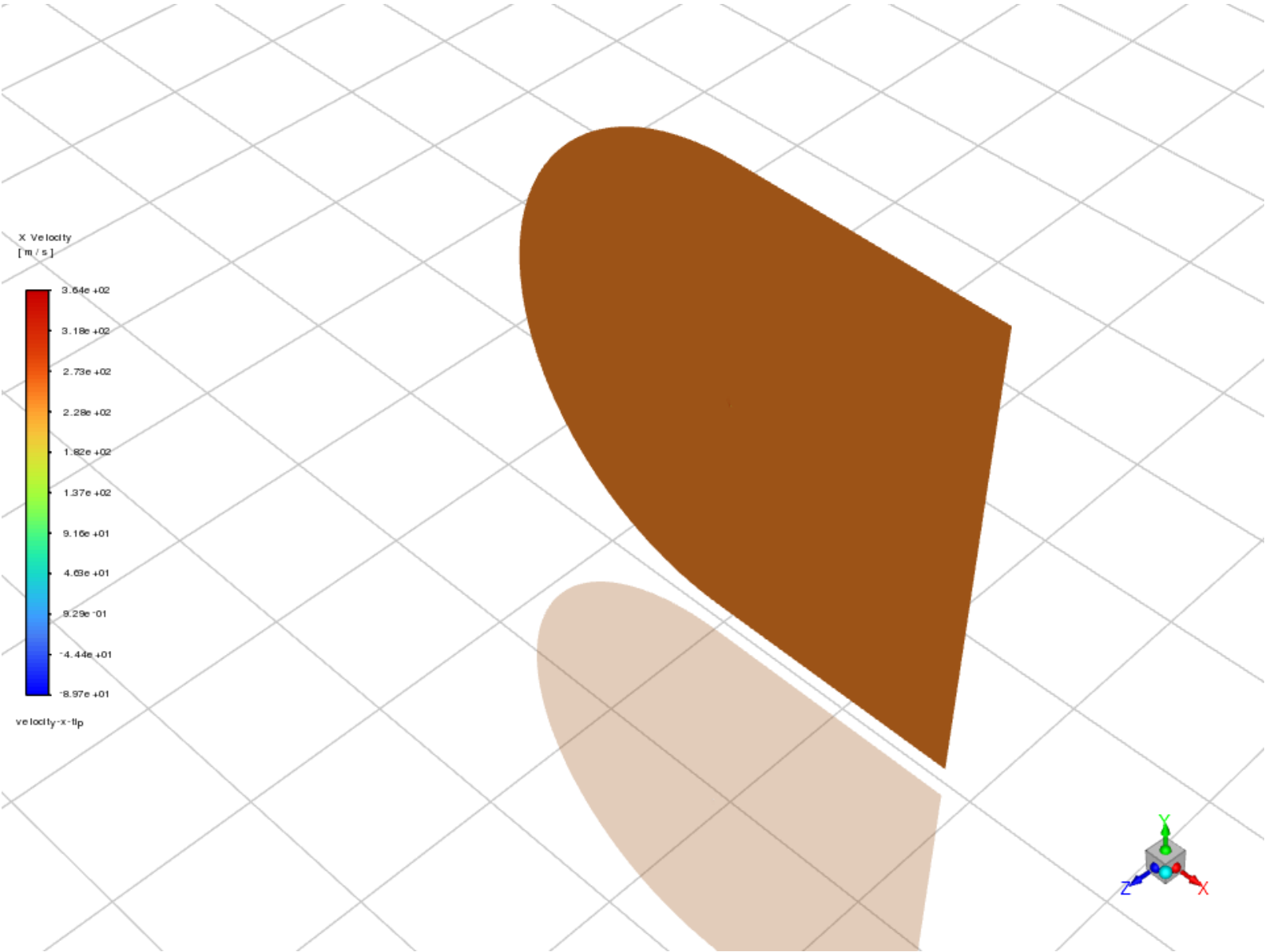
Plots

Residuals

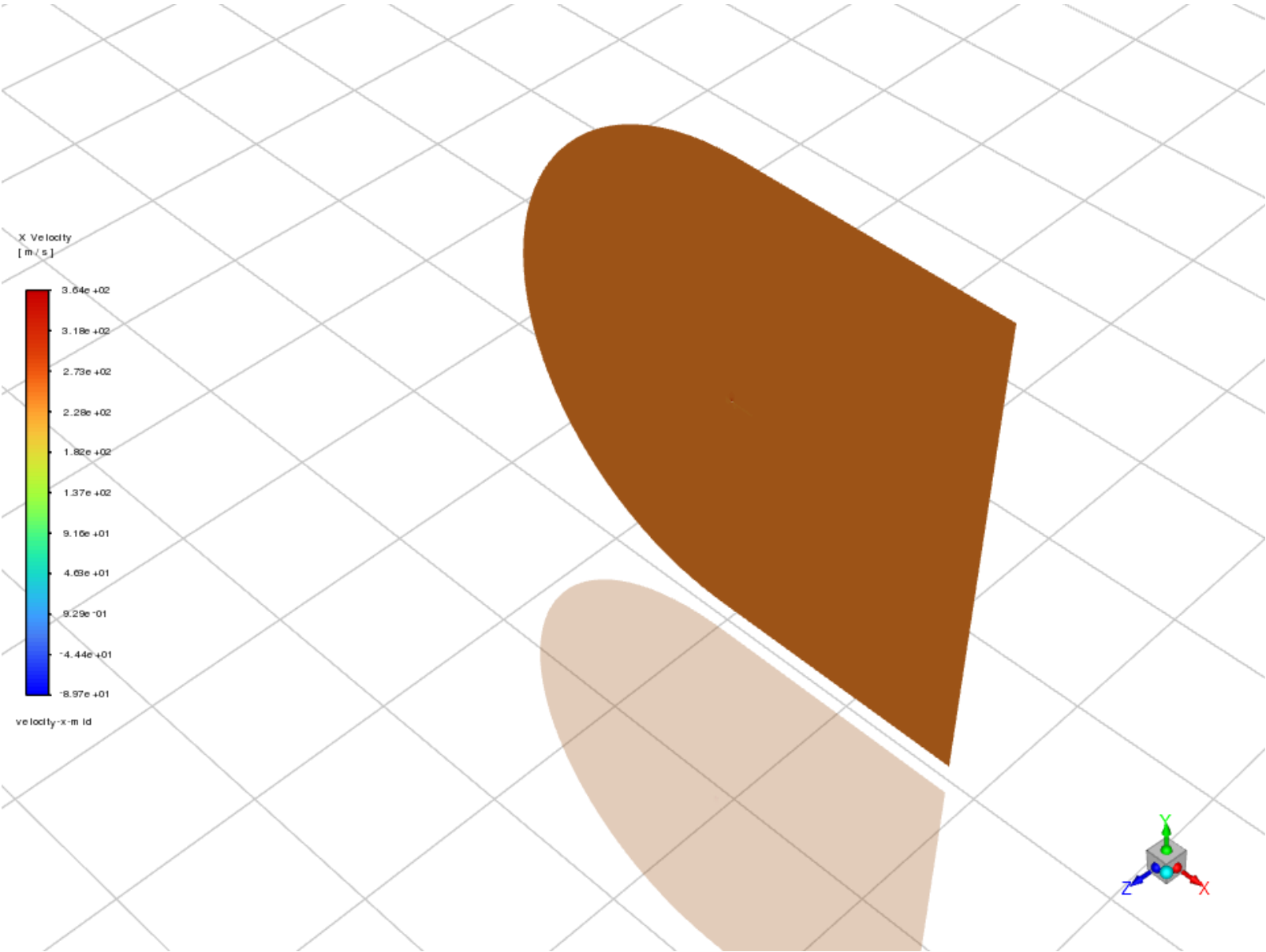


Contours

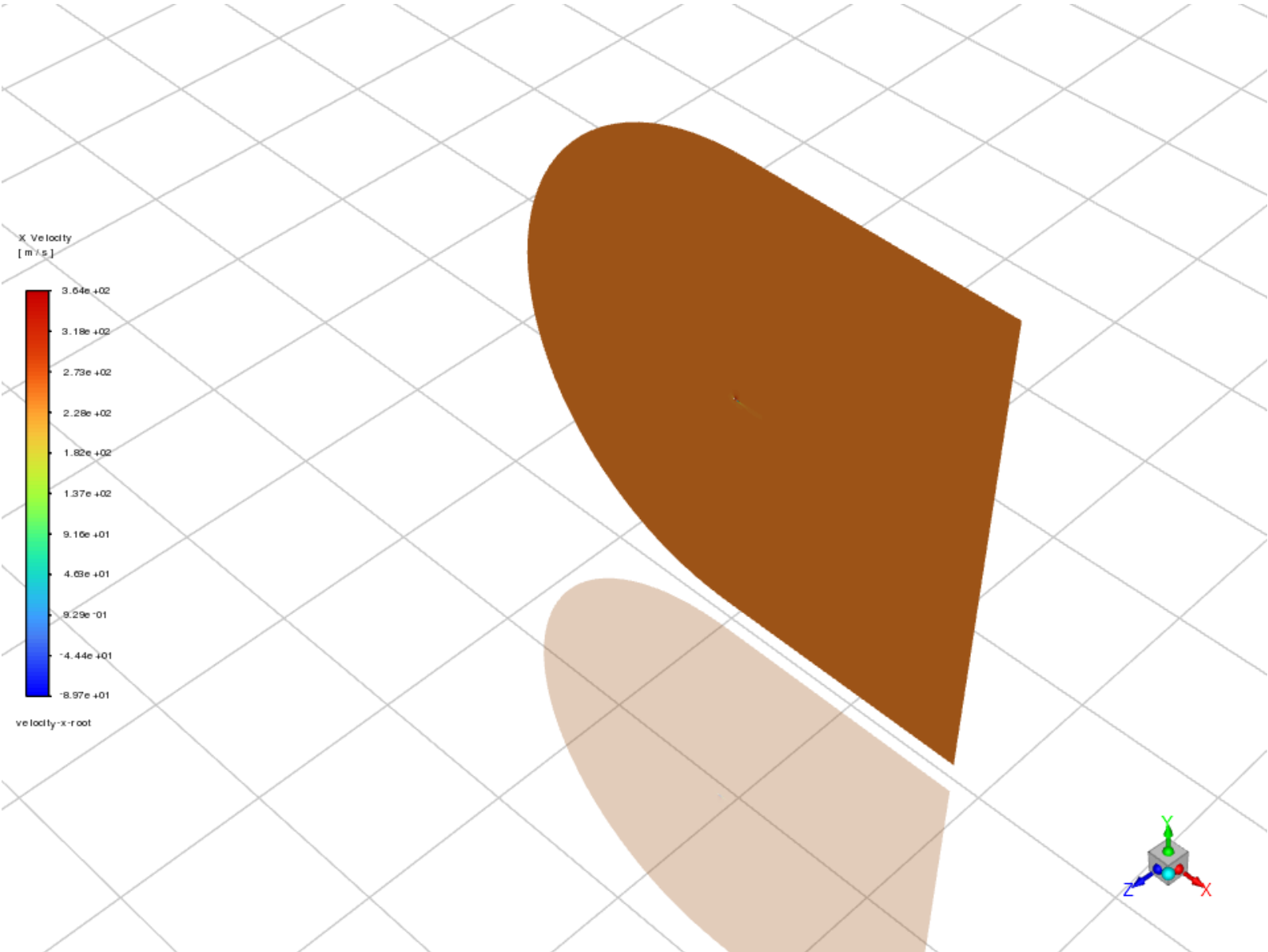
velocity-x-tip



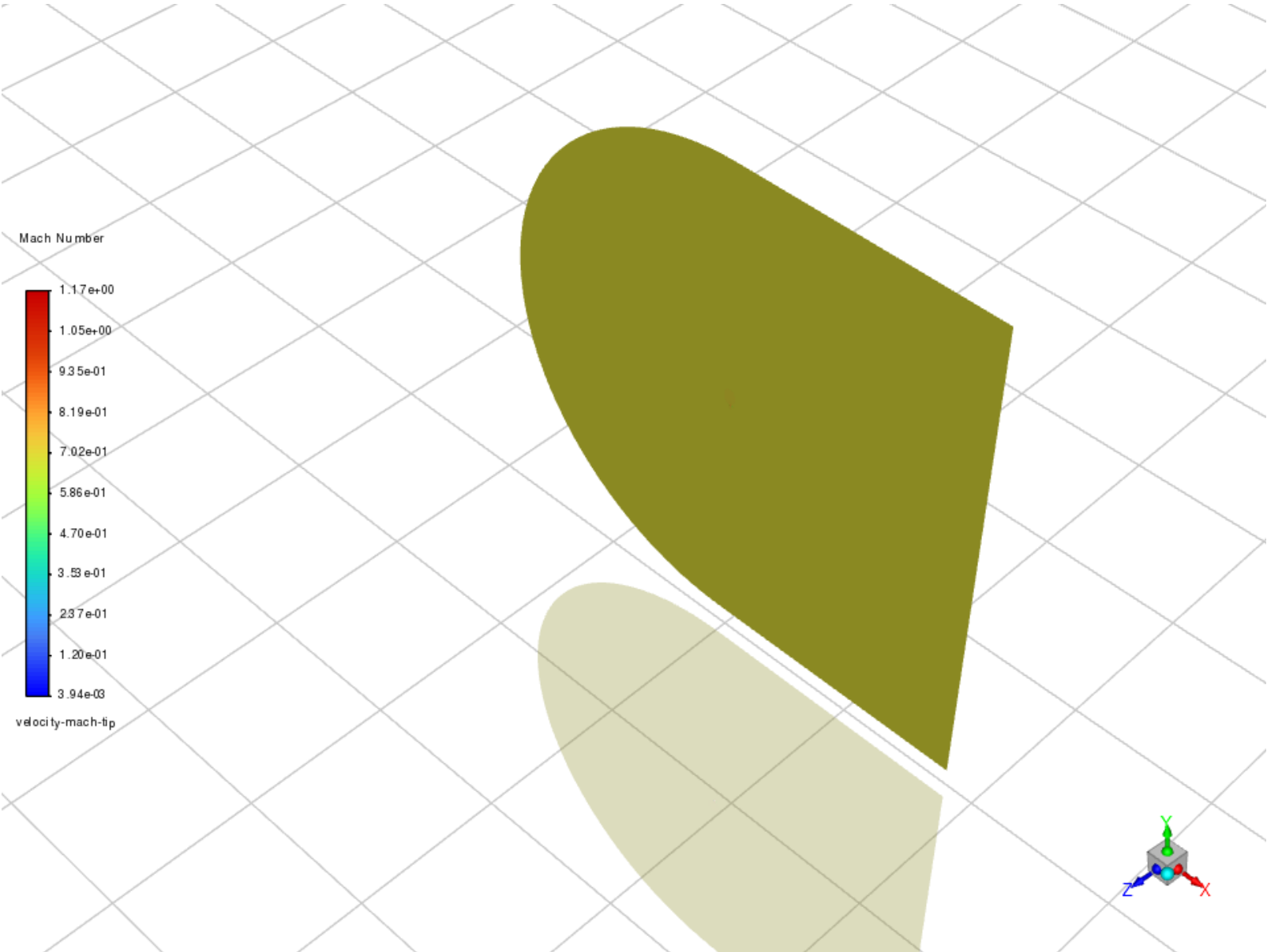
velocity-x-mid



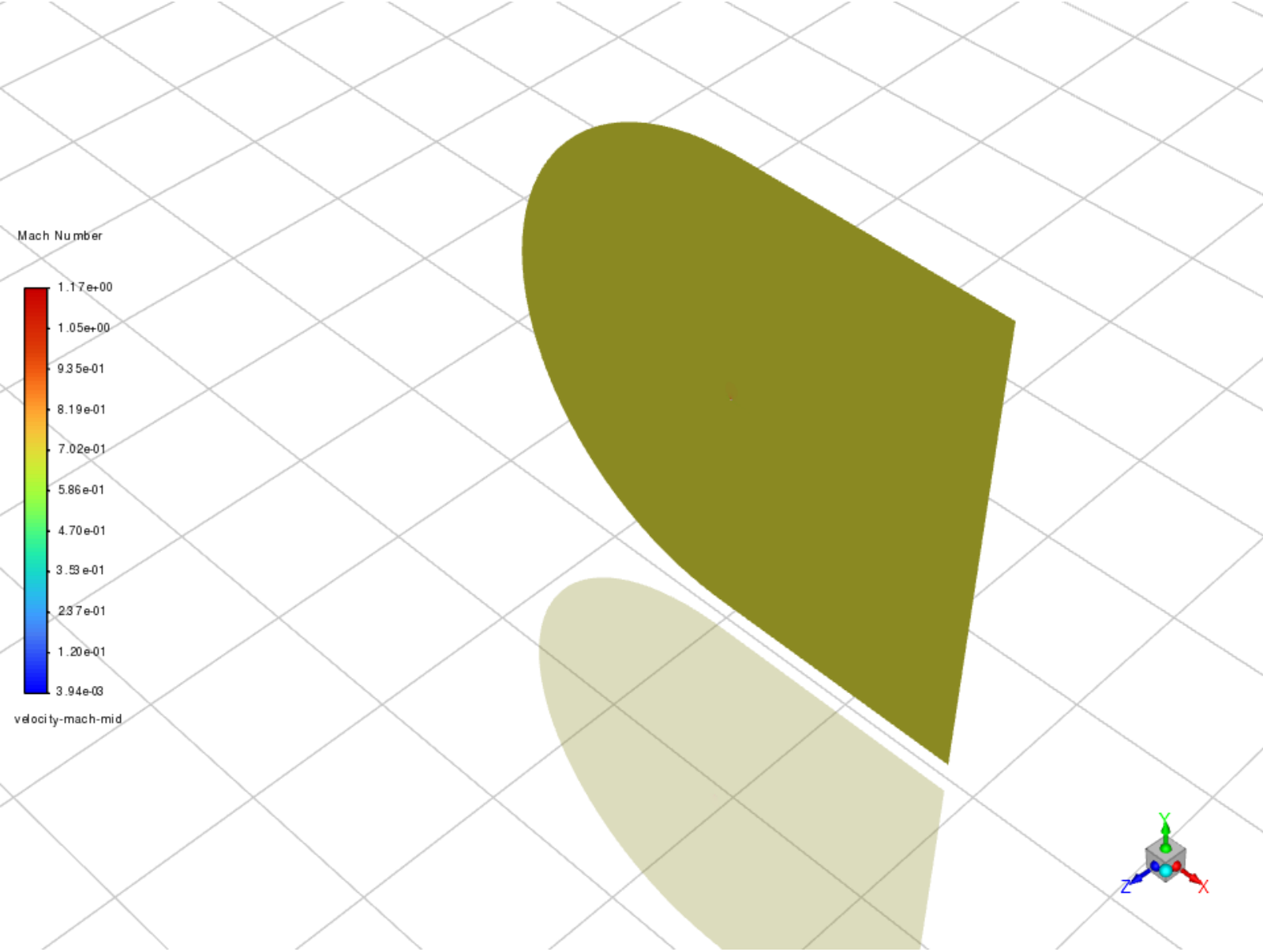
velocity-x-root



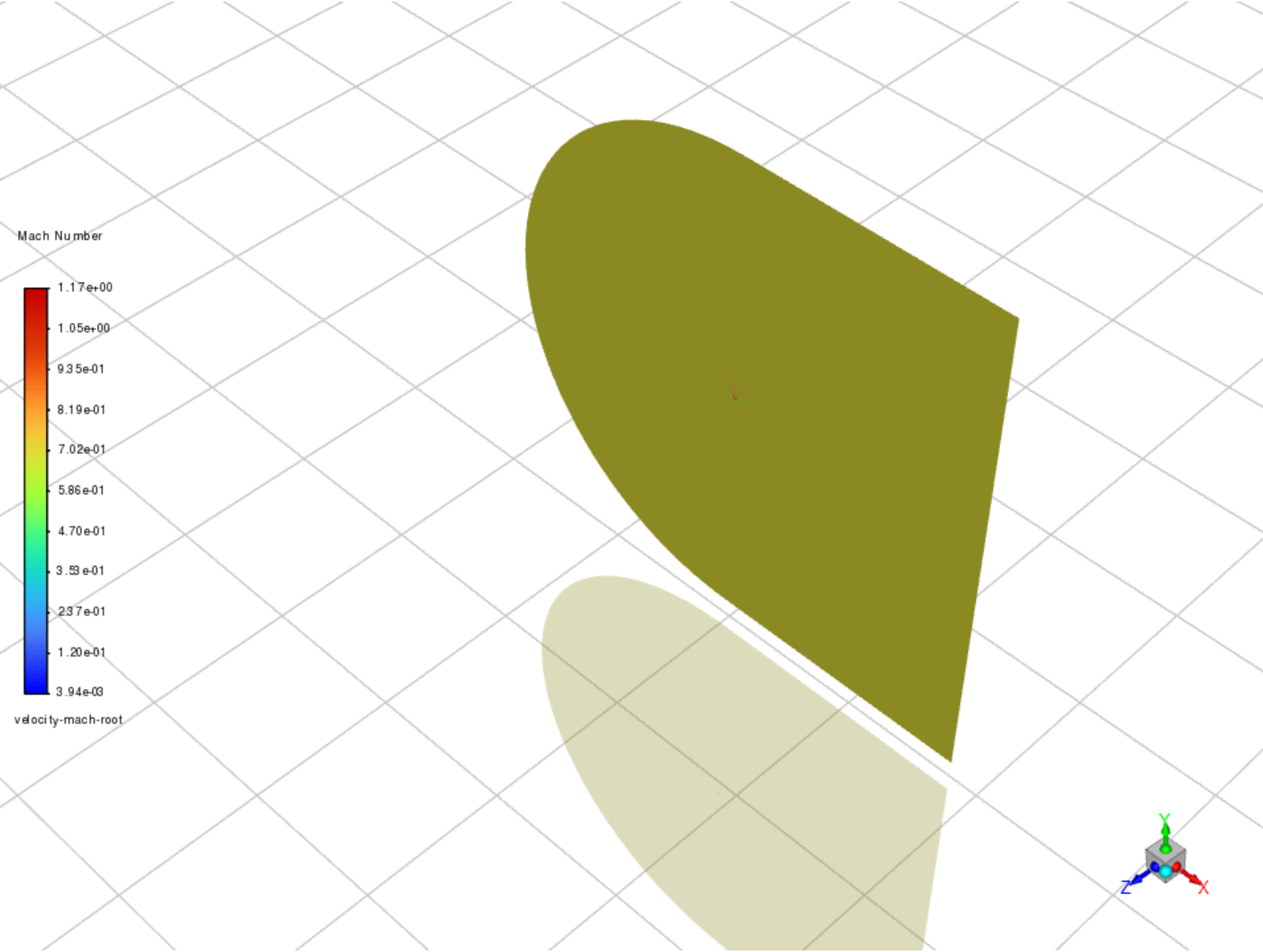
velocity-mach-tip



velocity-mach-mid

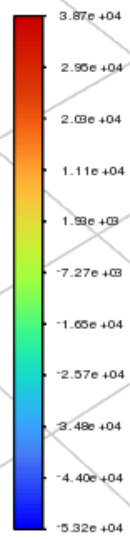


velocity-mach-root

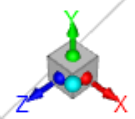


pressure

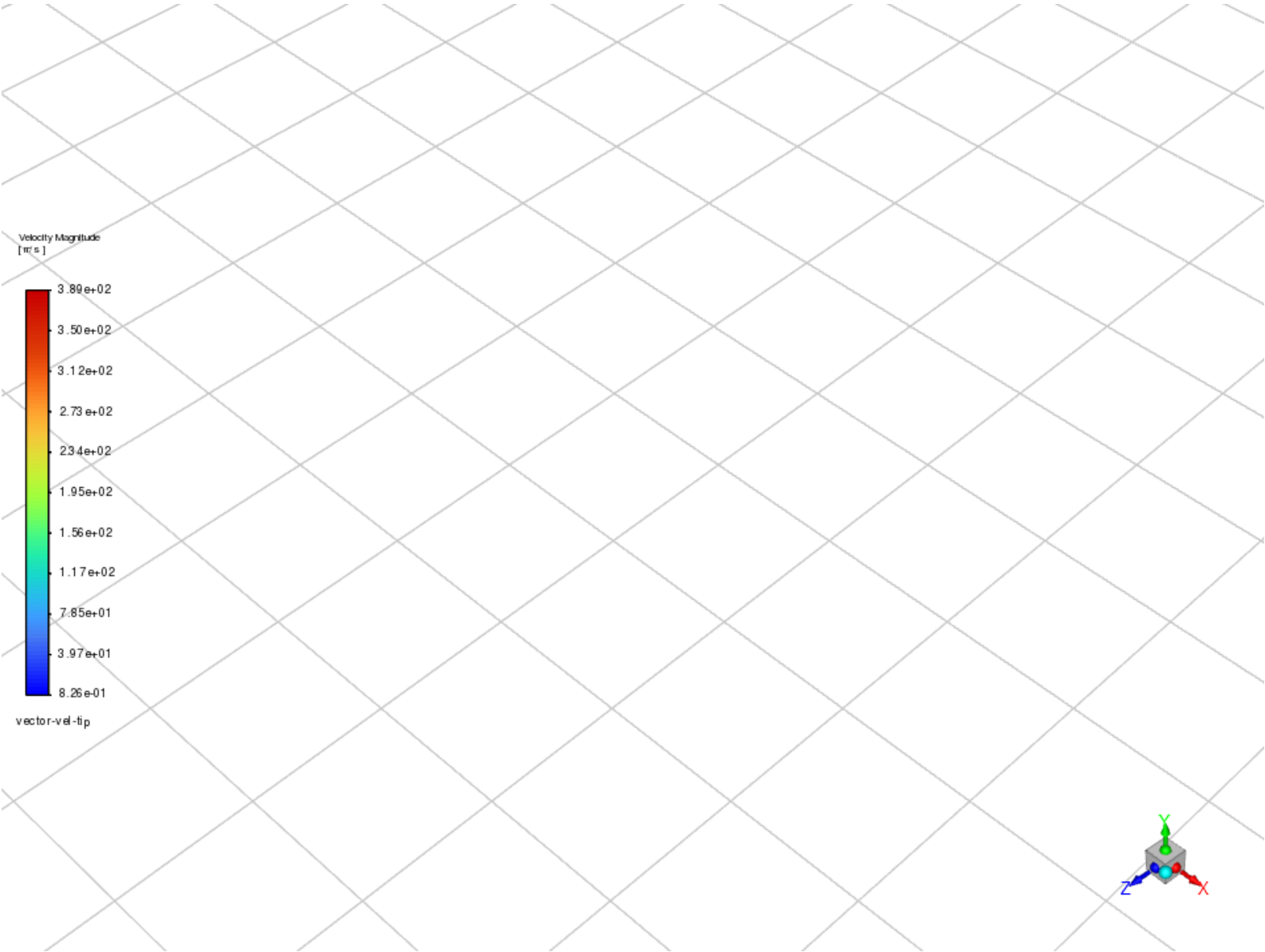
Static Pressure
[Pa]



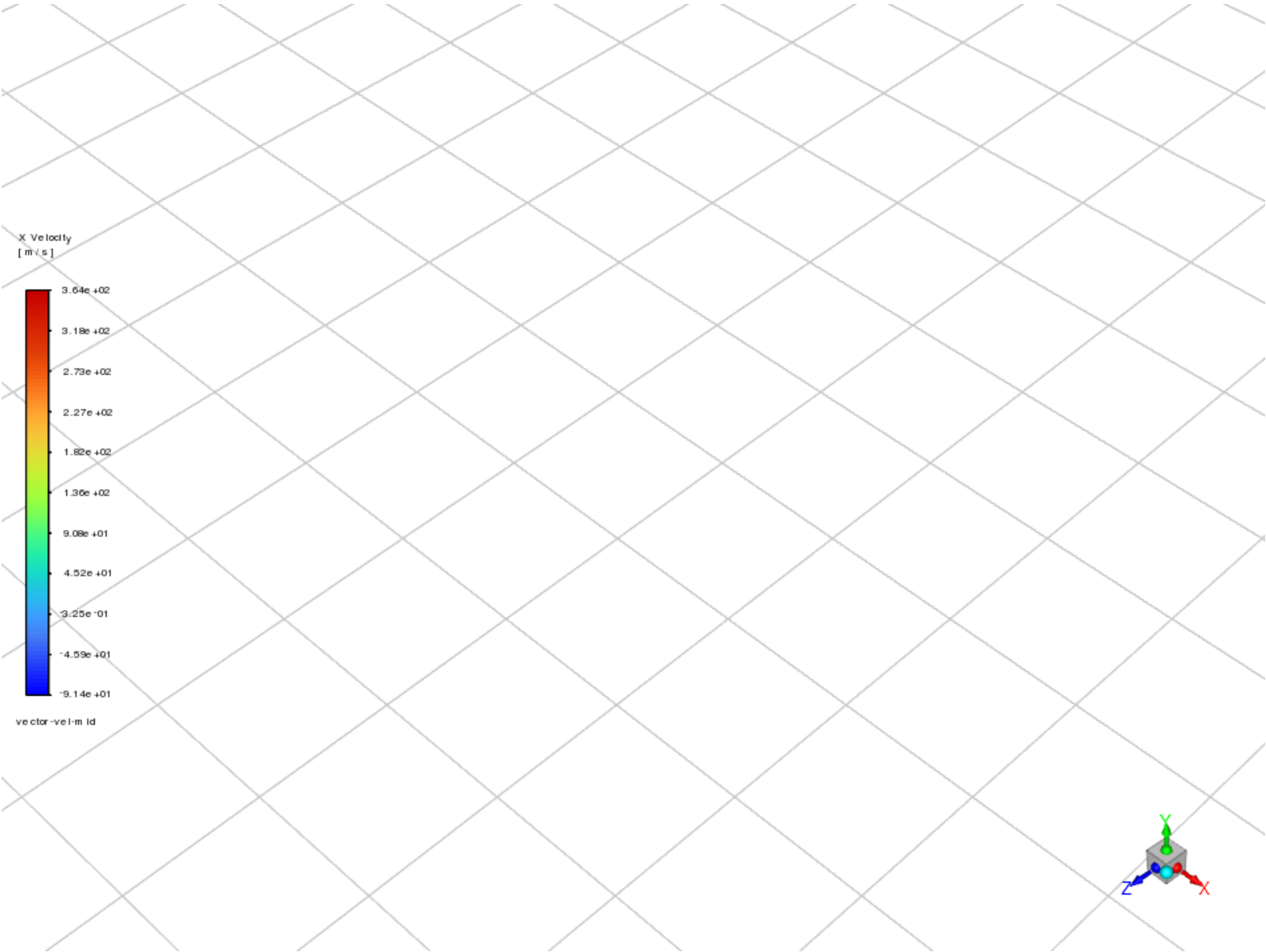
pressure



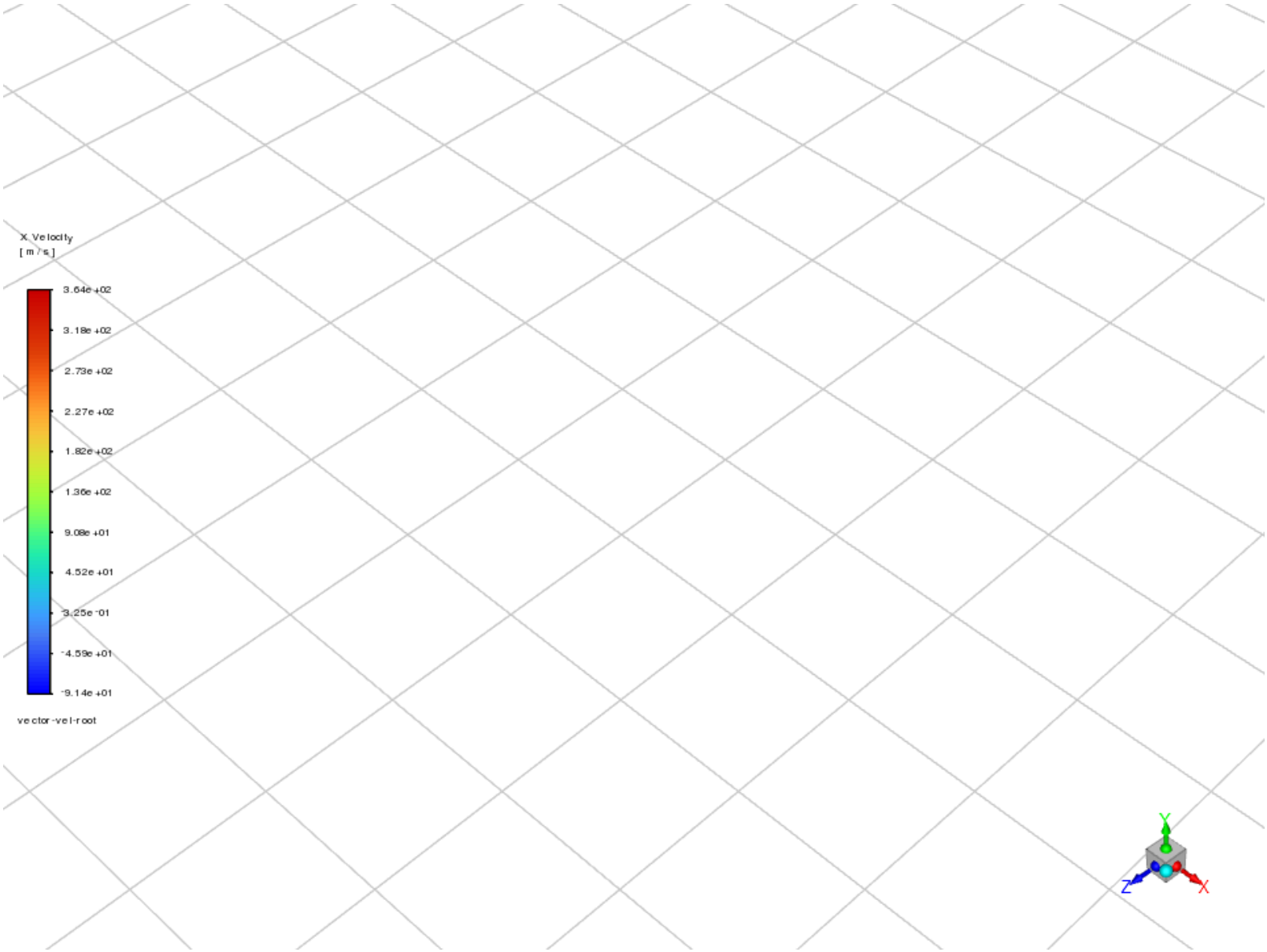
vector-vel-tip



vector-vel-mid



vector-vel-root



XY Plots

xy-x-shear-stress

Wall Shear Stress

